

**Due Date: 4/7/23**

Group 1 MATH 341

Homework 5

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Problem 1

- a) The underlying assumption behind Simpson's rule is that  $f(x)$  is four times continuously differentiable on the given interval  $[a, b]$ .

Handwritten derivation of the error term for Simpson's rule. The derivation shows the relationship between the error of Simpson's rule with  $n$  subintervals and with  $n/4$  subintervals. It uses the asymptotic error formula for Simpson's rule, which involves the fourth derivative of the function. The final result is a boxed 16, indicating the error approaches  $16h^4$ .

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- b) Based on the work from part a), we would expect the ratio  $R_n = \frac{I_{n/2} - I_{n/4}}{I_n - I_{n/2}}$  to approach 16

as  $n \rightarrow \infty$ . If we consider the number  $p_n = \frac{\log(R_n)}{\log(2)}$ , it is expected to approach a value of 4 as  $n \rightarrow \infty$  using logarithmic rules. Substituting 16 for  $R_n$  in the limit:

$$\lim_{n \rightarrow \infty} p_n = \frac{\log(16)}{\log(2)} = \frac{\log(2^4)}{\log(2)} = \frac{4 \log(2)}{\log(2)} = 4.$$

The base of the logarithmic function is 2,  $p_n$  is written above in the form of the change of

base formula ( $\log_b a = \frac{\log_d a}{\log_d b}$ ) and can be rewritten as

$$p_n = \log_2(16) = 4 * \log_2(2) = 4. \text{ Written in the form above the base is}$$

inconsequential. We can see from the change of base formula that as long as the bases are the same in both log functions, the value of  $d$  doesn't make a difference. For the purposes of MatLab, we want to use the  $\log_2(x)$  function to reduce computational complexity. In order to interpret this value in a practical situation, its motivation should be explained. It was assumed that the approximation has an error in the form  $E_n = Ch^p$ , for some constant  $C \neq 0$ , and  $p > 0$ . If  $p_n \rightarrow p$  as  $n \rightarrow \infty$ , the convergence order of the approximation can be estimated.

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Problem 2

The goal is to approximate the integrals  $I^{(1)} = \int_0^2 2x \sin x \, dx$ , and  $I^{(2)} = \int_0^4 \sin(\sqrt{x}) \, dx$

using Simpson's rule for several numbers of sub-intervals. This will be accomplished via Octave code. Additionally, we will examine the behavior of the ratio  $R_n$  and number  $p_n$  as defined in problem 1 as the value of  $n$  increases. Both of these values will be insightful toward the convergence interpretations of each integral approximation.

The results will be compared to calculations done by hand in order to check for programming accuracy. The true value of the integrals is not necessarily needed, as convergence of the approximations can be shown without it.

The following Octave code is in a separate file to the simpson function, and is used simply to define variables, format tables, and call the function to calculate the approximations.

```
1| # Alex Caramanico, March 30, 2023
2| fhandle1 = @(x)(2*x.*sin(x)); # f1(x)
3| a1 = 0; b1 = 2; i = 9; # lower/upper bound, number of different n values
4| disp('          I(1) = Integral of 2xsin(x) from 0 to 2') # disp() to label tables
5| disp(' n      Approximation      ln - ln/2      Rn      pn');
6| simpson(fhandle1,a1,b1,i); # call function with integrand, a, b, and i
7| fhandle2 = @(x)(sin(sqrt(x))); # f2(x)
8| a2 = 0; b2 = 4; # lower/upper bound, same number of n values
9| disp('          I(2) = Integral of sin(sqrt(x)) from 0 to 4')
10| disp(' n      Approximation      ln - ln/2      Rn      pn');
11| simpson(fhandle2,a2,b2,i)
```

The Octave code below is the function, defined in a separate file simpson.m, used to calculate the approximations and additional values of interest. It takes in the following parameters:

- $fx$  - the integrand function,
- $a$  - lower bound of integral,
- $b$  - upper bound of integral,
- $i$  - highest power to which 2 is to be raised (i.e. to determine the number of sub-intervals at each step)

For clarity, a brief explanation of the code is as follows. The function takes in the aforementioned parameters. The following variables are formatting strings for use in the fprintf function. Next, empty vectors are created to be used in calculations for the convergence columns in the table output. The for loop completes calculations for each value of sub-intervals  $n$ , which corresponds to the current approximation index. Next, the value for  $n$  is calculated, then the sub-interval width  $h$ , by which the vector of nodes is created. The following variables are for the indices of the node vector. It should be pointed out that Octave indexing starts at 1, so each *odd* functional value in Simpson's composite rule is a corresponding *even* index in the code. The inverse is also true. Each index vector value is spaced 2 from its neighbor(s), which effectively skips every other index. The even indices, represented by variable  $ind2$ , starting at 2 and ending at  $n$ , are multiplied by 4 in the approximation formula. The variables  $ind1$  and  $ind3$

have the effect of including the first and last node once, while doubling the odd nodes according to Simpson's composite rule. From here, the approximation (area) is calculated. The remainder of the code is simply for formatting the tables in the output. For reference, Matlab code would replace "endif", "endfor", or "endfunction" simply with "end". Otherwise, the code is identical.

```

1| function simpson(fx, a, b, i)
2| format1 = '%3i   %9.13e \n'; # formats for fprintf
3| format2 = '%3i   %9.13e   %9.13e \n';
4| format3 = '%3i   %9.13e   %9.13e   %9.13e   %9.13e \n';
5| areas = zeros(9,1); # vector for areas based on each n, for extra calculations
6| differences = zeros(9,1); # differences vector based on each n, avoid repeating calculations
7| for k=1:i; # for loop to make i approximations, each approximation with own n
8|     n = 2^k; # number of sub-intervals increases by factor of 2 with each approximation
9|     h = (b-a) / n; # sub-interval width
10|    x = a:h:b; # vector of equidistant nodes
11|    # odd nodes "doubled" through addition aside from 1st and last nodes
12|    ind1 = 1:2:n-1; ind2 = 2:2:n; ind3 = 1:2:n+1; # indices for each part of Simpson's
13|    current_area = (h/3)*(sum(fx(x(ind1))))+4*sum(fx(x(ind2)))+sum(fx(x(ind3)))); # Simpson's
14|    areas(k) = current_area; # update area vector for current approximation
15|    if k == 1; # conditionals to format table properly
16|        fprintf(format1,n,current_area) # print first approximation
17|    endif
18|    if k == 2;
19|        differences(k) = current_area-areas(k-1);
20|        fprintf(format2,n,current_area,differences(k)) # print all info, now have a difference
21|    endif
22|    if k >= 3;
23|        differences(k) = current_area-areas(k-1);
24|        R = differences(k-1) / differences(k); # Rn ratio
25|        p = log2(R); # pn number
26|        fprintf(format3,n,areas(k),differences(k),R,p) # print all info, now have ratio, pn
27|    endif
28| endfor
29| endfunction

```

For both tables, the columns represent the following:

1. Number of sub-intervals used in the approximation, increases by a factor of 2 each step
2. The value of the approximation itself, represented to ten decimal places
3. Difference between the current approximation and the previous approximation
4. Ratio between the current difference and the previous difference
5. Log base 2 of the ratio, a value used to approximate the convergence order  $p$  as  $n \rightarrow \infty$ .

This is used for the interpretation of the error assumption  $E_n = Ch^p$

The following table was produced by the code for the first integral  $I^{(1)}$ :

| I (1) = Integral of 2xsin(x) from 0 to 2 |                  |                  |            |            |
|------------------------------------------|------------------|------------------|------------|------------|
| n                                        | In               | In - In/2        | Rn         | pn         |
| 2                                        | 3.4563191953e+00 |                  |            |            |
| 4                                        | 3.4817859400e+00 | 2.5466744778e-02 |            |            |
| 8                                        | 3.4830987143e+00 | 1.3127742840e-03 | 1.9399e+01 | 4.2779e+00 |
| 16                                       | 3.4831770389e+00 | 7.8324595024e-05 | 1.6761e+01 | 4.0670e+00 |
| 32                                       | 3.4831818782e+00 | 4.8392508689e-06 | 1.6185e+01 | 4.0166e+00 |
| 64                                       | 3.4831821797e+00 | 3.0158572306e-07 | 1.6046e+01 | 4.0041e+00 |
| 128                                      | 3.4831821986e+00 | 1.8835583759e-08 | 1.6011e+01 | 4.0010e+00 |
| 256                                      | 3.4831821998e+00 | 1.1770127095e-09 | 1.6003e+01 | 4.0003e+00 |
| 512                                      | 3.4831821998e+00 | 7.3561601255e-11 | 1.6000e+01 | 4.0000e+00 |

The approximation appears to be approaching a value near 3.4831821998 as  $n$  increases. This can be said because each successive step has an approximation with more decimal places in agreement with the last. It can be clearly seen that the difference between approximations  $I_n - I_{n/2}$  approaches 0 as  $n \rightarrow \infty$ , as the number of 0's in front of the calculation is increasing.

The ratio  $R_n$  is approaching 16 as  $n \rightarrow \infty$ , thereby the number  $p_n \rightarrow 4$  as  $n \rightarrow \infty$ .

Convergence of the approximation can be concluded based on the above data, primarily because the difference between successive approximations is approaching 0. This essentially means that the algorithm is becoming closer and closer to a singular value - that being the true value of the integral. Additionally, the ratio  $R_n$  approaches 16 and  $p_n$  approaches 4 as  $n$  increases, both of which follow the expected convergence values shown in the first problem.

The table below was produced by the code for the second integral  $I^{(2)}$ :

| I (2) = Integral of sin(sqrt(x)) from 0 to 4 |                  |                  |            |            |
|----------------------------------------------|------------------|------------------|------------|------------|
| n                                            | In               | In - In/2        | Rn         | pn         |
| 2                                            | 3.2402408072e+00 |                  |            |            |
| 4                                            | 3.3996066127e+00 | 1.5936580547e-01 |            |            |
| 8                                            | 3.4540572386e+00 | 5.4450625903e-02 | 2.9268e+00 | 1.5493e+00 |
| 16                                           | 3.4729599698e+00 | 1.8902731252e-02 | 2.8806e+00 | 1.5264e+00 |
| 32                                           | 3.4795812656e+00 | 6.6212957527e-03 | 2.8548e+00 | 1.5134e+00 |
| 64                                           | 3.4819113907e+00 | 2.3301251017e-03 | 2.8416e+00 | 1.5067e+00 |
| 128                                          | 3.4827333082e+00 | 8.2191747446e-04 | 2.8350e+00 | 1.5033e+00 |
| 256                                          | 3.4830235645e+00 | 2.9025631132e-04 | 2.8317e+00 | 1.5017e+00 |
| 512                                          | 3.4831261264e+00 | 1.0256197657e-04 | 2.8301e+00 | 1.5008e+00 |

As  $n$  increases, the approximation appears to converge to a value near 3.4831261264. Further, the difference between approximations appears to be converging to 0. The ratio  $R_n$  is nearing a value near 2.8301, while the number  $p_n$  is approaching a value near 1.5. Aside from the approximation value (especially for a greater number of sub-intervals) these results are in strong contrast to the results from  $I^{(1)}$ .

To gain a complete understanding of the algorithm, one might compare the behavior between both integrals. Though both have improving approximations for larger values of  $n$ , there are various differences that should be analyzed. It can be seen that the approximation for  $I^{(1)}$  converges more quickly, as its differences at each step are significantly lower than the differences at each corresponding step for  $I^{(2)}$ . For  $n = 512$ , the differences are separated by a

factor of  $10^7$ . The difference in convergence speed could also explain the dramatically different results for the ratio  $R_n$  and corresponding value  $p_n$ . Since the integrand leads to extremely slow convergence, the errors decrease at a smaller rate, causing smaller values for the ratio, thus smaller values for the logarithmic number.

A similarity that could be mentioned is that both integrals start as under-estimates, primarily because each integrand is increasing across most  $(2x\sin x)$  or all  $(\sin\sqrt{x})$  of their intervals.

The results may vary in the ways discussed as a result of the asymptotic error theorem for numerical integration. Based upon the original error theorem:  $E_n(f) = \frac{-h^4(b-a)}{180}f^{(4)}(\xi)$ , the

asymptotic formula  $E_n^{\sim}(f)$  can be derived. The idea of the following is to remove the dependence upon the intermediate value  $\xi$ . The following can be shown using the definition of a definite integral. The key assumption of the composite Simpson rule is that  $f(x)$  is four times continuously differentiable on the interval  $[a, b]$

$$E_n(f) \approx E_n^{\sim}(f) = \frac{-h^4}{180}(f'''(b) - f'''(a))$$

As seen above, the asymptotic error theorem involves the third and fourth derivatives of the integrand. The first derivatives are below to demonstrate the issue with the second integral:

for  $I^{(1)}$ :  $f_{(1)}'(x) = 2\sin(x) + 2x\cos(x)$ , and

for  $I^{(2)}$ :  $f_{(2)}'(x) = \frac{\cos(\sqrt{x})}{2\sqrt{x}}$ .

It is clear that the first (and by extension second through fourth) derivative for  $f_{(2)}(x)$  is asymptotic at  $x = 0$  as a result of the chain rule. Both integrals have lower bound  $a = 0$ , the problem is clearly seen then because the derivatives of the second integrand (from  $I^{(2)}$ ) are undefined at this value. In fact, if the limit was taken to approximate the value, it can be seen that the derivatives diverge to  $\infty$  as  $x \rightarrow 0^+$ . This in turn leads to an infinite value for the asymptotic error theorem. Ultimately, the derivatives of the second integrand are not *continuously* differentiable on the interval  $[a, b]$ , because they have an asymptote at  $x = a = 0$ . This is in violation of the fundamental assumption behind Simpson's composite rule - that the function  $f(x)$  is four times continuously differentiable on the interval  $[a, b]$ . For this reason, it would be significantly more reliable to use the approximation based around  $I^{(1)}$ , since it has stronger convergence properties by satisfying the fundamental assumption. In a practical setting, it might be reported that the true value of the integral is 3.4831821998, with higher precision possible using different Octave decimal truncation.

Lastly, it is useful to check the accuracy of the computer results by completing the calculations by hand. Below are images of these hand-calculations. The only "inaccuracies" are due to truncating the approximations at six decimal places, leading to very slight differences for  $R_n$  and  $p_n$ . Otherwise, the results are exactly the same.

HW 5 H2 Approx (Crammer)

$$I_n(f) = \frac{h}{3} [f_0 + 4f_1 + 2f_2 + 4f_3 + \dots + 2f_{n-2} + 4f_{n-1} + f_n], R_n = \frac{I_{n/2} - I_n}{I_n - I_{n/2}}, \rho_n = \frac{1/2 R_n}{1/2 R_n}$$

$$I^{(1)} = \int_0^2 2x \sin x \, dx$$

$n=2$ :  $h=1 \rightarrow x = [0, 1, 2]$   $I_2(f) = \frac{1}{3} [2(0) \sin(0) + 4 \cdot 2(1) \sin(1) + 2(2) \sin(2)] = 3.456319$

$n=4$ :  $h = \frac{1}{2} \rightarrow x = [0, \frac{1}{2}, 1, \frac{3}{2}, 2]$   
 $I_4(f) = \frac{1}{6} [2(0) \sin(0) + 4 \cdot 2(\frac{1}{2}) \sin(\frac{1}{2}) + 2 \cdot 2(1) \sin(1) + 4 \cdot 2(\frac{3}{2}) \sin(\frac{3}{2}) + 2(2) \sin(2)] = 3.481785$

$n=8$ :  $h = \frac{1}{4} \rightarrow x = [0, \frac{1}{4}, \frac{2}{4}, \frac{3}{4}, \frac{4}{4}, \frac{5}{4}, \frac{6}{4}, \frac{7}{4}, 2]$   
 $I_8(f) = \frac{1}{12} [2(0) \sin(0) + 4 \cdot 2(\frac{1}{4}) \sin(\frac{1}{4}) + 2 \cdot 2(\frac{2}{4}) \sin(\frac{2}{4}) + 4 \cdot 2(\frac{3}{4}) \sin(\frac{3}{4}) + 2 \cdot 2(1) \sin(1) + 4 \cdot 2(\frac{5}{4}) \sin(\frac{5}{4}) + 2 \cdot 2(\frac{6}{4}) \sin(\frac{6}{4}) + 4 \cdot 2(\frac{7}{4}) \sin(\frac{7}{4}) + 2(2) \sin(2)] = 3.483098$

| n | approx   | $I_n - I_{n/2}$ | $R_n$  | $\rho_n$ |
|---|----------|-----------------|--------|----------|
| 2 | 3.456319 | N/A             | N/A    | N/A      |
| 4 | 3.481785 | 0.025466        | N/A    | N/A      |
| 8 | 3.483098 | 0.001313        | 14.395 | 4.277    |

all results match  
programming results

$$I^{(2)} = \int_0^4 \sin(\sqrt{x}) \, dx$$

$n=2$ :  $\rightarrow h=2 \rightarrow x = [0, 2, 4]$ ,  $I_2(f) = \frac{2}{3} [\sin(\sqrt{0}) + 4 \sin(\sqrt{2}) + \sin(\sqrt{4})] = 3.240240$

$n=4$ :  $\rightarrow h=1 \rightarrow x = [0, 1, 2, 3, 4]$ ,  $I_4(f) = \frac{1}{3} [\sin(\sqrt{0}) + 4 \sin(\sqrt{1}) + 2 \sin(\sqrt{2}) + 4 \sin(\sqrt{3}) + \sin(\sqrt{4})] = 3.391606$

$n=8$ :  $\rightarrow h = \frac{1}{2} \rightarrow x = [0, \frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}, 3, \frac{7}{2}, 4]$ ,  
 $I_8(f) = \frac{1}{6} [\sin(\sqrt{0}) + 4 \sin(\sqrt{\frac{1}{2}}) + 2 \sin(\sqrt{1}) + 4 \sin(\sqrt{\frac{3}{2}}) + 2 \sin(\sqrt{2}) + 4 \sin(\sqrt{\frac{5}{2}}) + 2 \sin(\sqrt{3}) + 4 \sin(\sqrt{\frac{7}{2}}) + \sin(\sqrt{4})] = 3.454057$

| n | approx   | $I_n - I_{n/2}$ | $R_n$    | $\rho_n$ |
|---|----------|-----------------|----------|----------|
| 2 | 3.240240 | -               | -        | -        |
| 4 | 3.391606 | 0.151366        | -        | -        |
| 8 | 3.454057 | 0.062451        | 2.126778 | 1.841303 |

all results match  
programming results

As seen in the previous results, it is clear that the two integrals have the same value. This can be shown through a transformation/substitution.

$$I^{(1)} = \int_0^2 2x \sin x \, dx, I^{(2)} = \int_0^4 \sin(\sqrt{x}) \, dx$$

For  $I^{(2)}$ , let  $u = \sqrt{x}$ , then  $du = \frac{1}{2\sqrt{x}} dx \leftrightarrow dx = 2\sqrt{x} du = 2u du$ .

To change the bounds of the integral, substitute  $x = 0$ , and  $x = 4$  (the original bounds) for the u-substitution:  $u = 0$ ,  $u = 2$ , respectively. Then compile the results into the new integral by rewriting everything in the terms as defined above:

$$I^{(2)} = \int_0^2 2u \sin(u) du, \text{ which is equivalent to } I^{(1)} = \int_0^2 2x \sin(x) dx \text{ upon evaluation. In practice, the}$$

variable name does not affect the value of the integral, therefore the two are equivalent.

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